

DS325 Final Project Report

Credit Card Fraud Detection Using Machine Learning

Ha Phung – Spring 2025

Professor Eatai Roth – Applied Data Science (DS325)

1. Introduction

Credit card fraud is a serious problem that can cause major financial losses for both customers and businesses. Most traditional systems rely on fixed rules or manual review to catch fraud, but these methods don't scale well as online transactions keep growing. In this project, we explore whether machine learning can help detect fraud automatically by learning patterns from past transaction data.

We used a real dataset of over 284,000 credit card transactions, including a small number of known fraud cases. Our goal was to build models that can predict whether a transaction is fraudulent or not, even though fraud cases are very rare in the data. We trained two machine learning models—Logistic Regression and Random Forest—on both an undersampled dataset and a dataset balanced using SMOTE, a technique that generates synthetic fraud cases.

To see how well the models worked, we looked at metrics like precision, recall, F1 score, and ROC AUC. Our results show that Random Forest trained on SMOTE data gave the best performance, especially in identifying fraud without too many false alarms.

We also used ChatGPT at one point in the project to help figure out how to apply SMOTE correctly and how to plot ROC curves for comparing models.

2. Methods

We used the Credit Card Fraud Detection dataset from Kaggle, which includes 284,807 transactions made in Europe. Each transaction is labeled as either legitimate (class 0) or fraudulent (class 1). The dataset includes 30 features: 28 anonymized numerical features from a PCA transformation (V1 to V28), plus Time (seconds since the first transaction) and Amount (the transaction value). The target variable is Class.

The first step was to clean and prepare the data. We found no missing values. The PCA features were already scaled, but Time and Amount were not, so we standardized those two columns using StandardScaler. We then split the data into training and test sets using stratified sampling to preserve the proportion of fraud cases (which make up only 0.17% of the data).

To handle the severe class imbalance in the training set, we used two different strategies:

- **Undersampling**, where we randomly reduced the number of non-fraud cases to match the number of fraud cases.
- **SMOTE (Synthetic Minority Oversampling Technique)**, which creates synthetic fraud samples to balance the dataset without removing any legitimate transactions.

We trained two classification models on both versions of the training data:

- **Logistic Regression**, a simple and interpretable linear model.
- **Random Forest**, an ensemble model based on decision trees.

We used scikit-learn for all model training and evaluation. The models were evaluated on the original (imbalanced) test set using precision, recall, F1 score, and ROC AUC.

3. Results

We evaluated each model using the imbalanced test set, which better reflects real-world conditions. The results focused on how well each model could identify fraud cases without producing too many false positives.

Logistic Regression trained on the undersampled data achieved decent precision but lower recall, which means it caught fewer fraud cases. Random Forest performed slightly better, with improved recall and F1 score. However, both models had limited access to the full training data since undersampling removes many legitimate examples.

When we used SMOTE to generate synthetic fraud cases instead of removing data, performance improved across the board. Logistic Regression trained on SMOTE data had higher recall and F1 score than its undersampled version, while Random Forest trained on SMOTE gave the best overall performance. It achieved the highest recall and F1 score, and its ROC AUC was around 0.97, showing strong ability to separate fraud from legit transactions.

The confusion matrices also showed that SMOTE-trained models caught more fraud cases without dramatically increasing false positives. The ROC curves confirmed that Random Forest with SMOTE had the best balance between sensitivity and specificity.

In summary, models trained on SMOTE-augmented data were more effective than those trained on undersampled data, and Random Forest was the best-performing algorithm overall.

4. Discussion

This project showed that machine learning models can be effective in detecting credit card fraud, especially when trained on balanced data. One of the biggest challenges was the extreme class imbalance in the dataset. With only 0.17% of the transactions labeled as fraud, a model could easily achieve high accuracy by predicting everything as legitimate. That's why we focused on precision, recall, and F1 score rather than accuracy.

Undersampling helped balance the classes but had limitations. By removing a large portion of legitimate transactions, we risked throwing away useful patterns and making the model less generalizable. While this approach was simple and allowed for faster training, it didn't make the best use of the data.

SMOTE was a better solution in this case. It allowed us to keep all of the original data while adding synthetic fraud samples to balance the training set. The models trained with SMOTE, especially the Random Forest, performed better in detecting fraud while keeping false positives at a reasonable level. The recall and F1 score were noticeably higher, and the ROC curve showed strong separation between the two classes.

One thing we might have improved was model tuning. We used default settings for both classifiers, but adjusting hyperparameters (like the number of trees in the Random Forest or the regularization strength in Logistic Regression) could further improve results. We also considered other models like Gradient Boosted Trees but decided to stick with the classics.

This project suggests that SMOTE and Random Forest together form a strong approach to fraud detection. In the future, we could explore real-time fraud prediction systems or cost-sensitive learning, where false negatives are penalized more heavily than false positives.

5. Citations

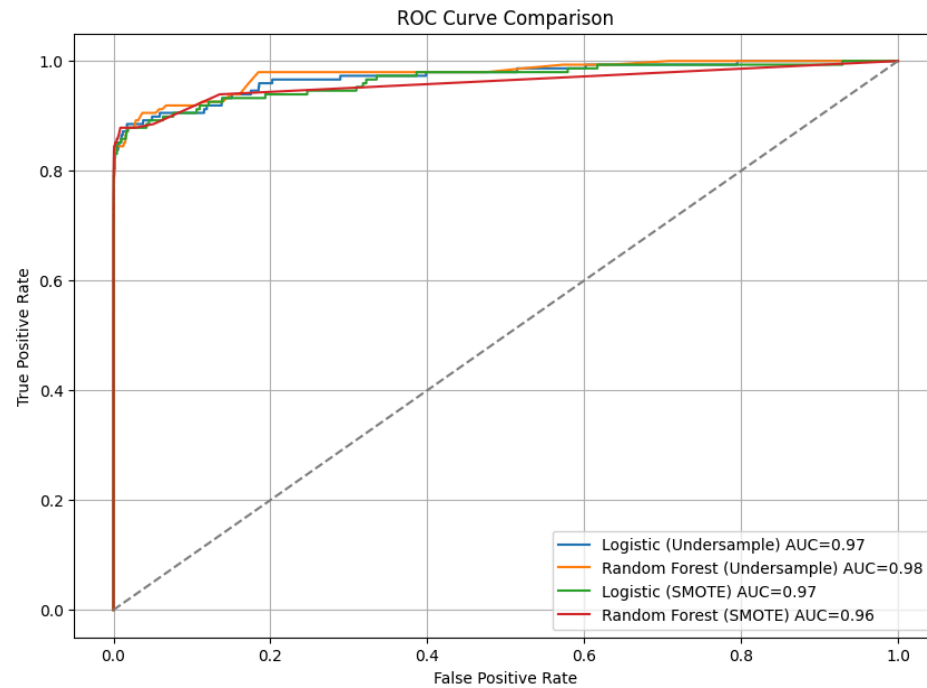
- Dal Pozzolo, A., Caelen, O., Johnson, R. A., & Bontempi, G. (2015). *Calibrating Probability with Undersampling for Unbalanced Classification*. In IEEE Symposium Series on Computational Intelligence (pp. 159-166).
- Dataset URL: <https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>
- Generative AI tools: ChatGPT by OpenAI, used to clarify implementation of SMOTE in scikit-learn and how to compare models using ROC curves.

6. Figures and Captions

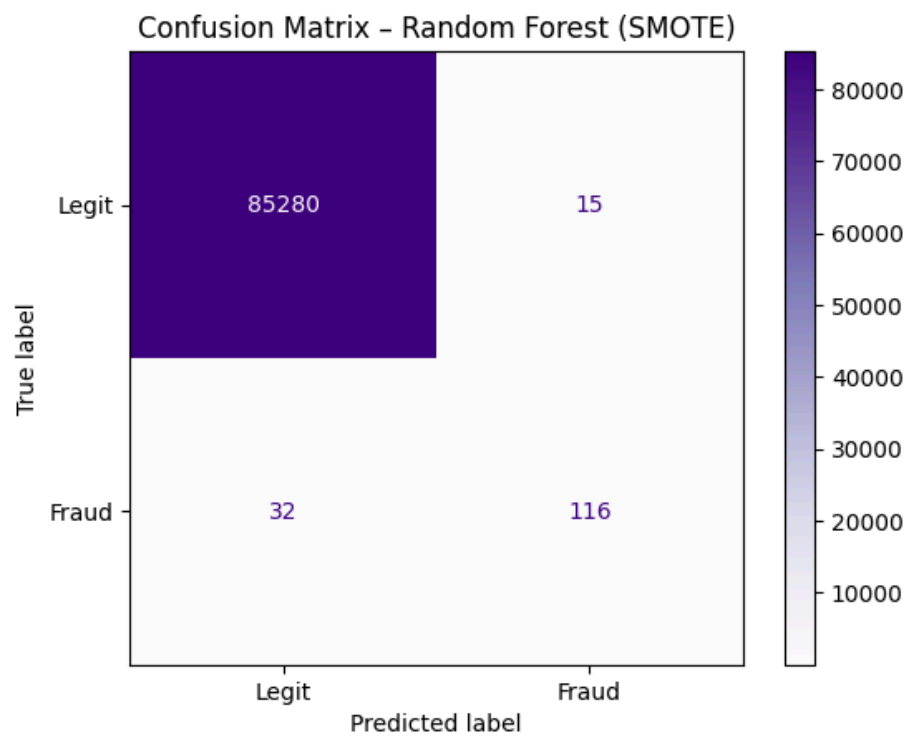
- **Figure 1:** Class distribution (Fraud vs. Legit)



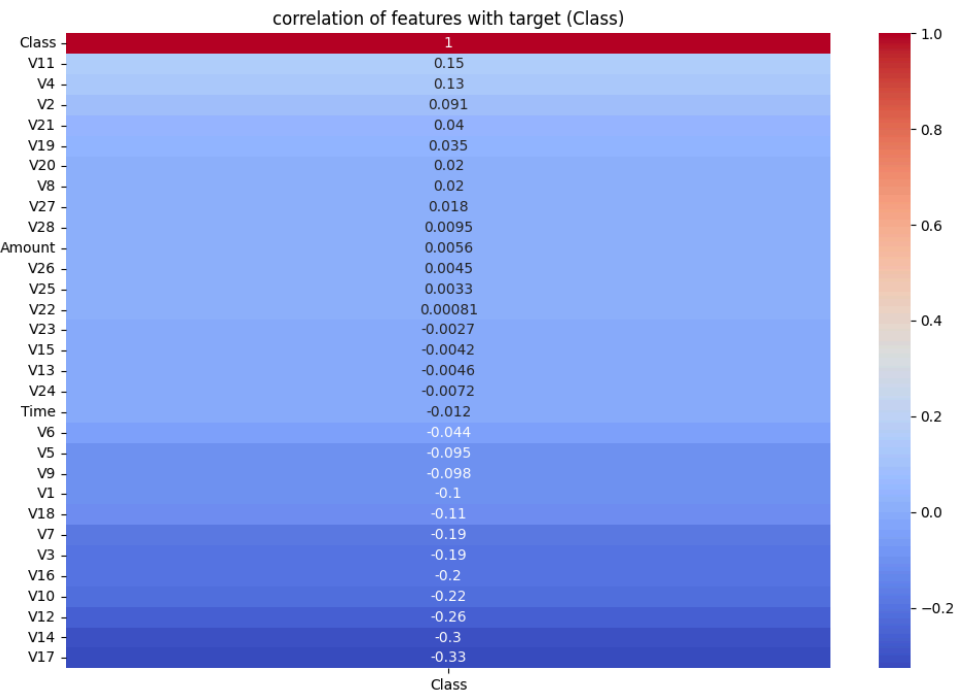
- **Figure 2:** ROC Curve comparison of all models



- **Figure 3:** Confusion Matrix (Random Forest with SMOTE)



- **Figure 4:** Feature boxplots (e.g., V14, V17) by Class



- **Table 1:** Precision, Recall, and F1 scores for all 4 models

	precision	recall	f1
Logistic (undersample)	0.063996	0.885135	0.119362
Random Forest (undersample)	0.078659	0.871622	0.144295
Logistic (SMOTE)	0.063446	0.878378	0.118343
Random Forest (SMOTE)	0.885496	0.783784	0.831541